

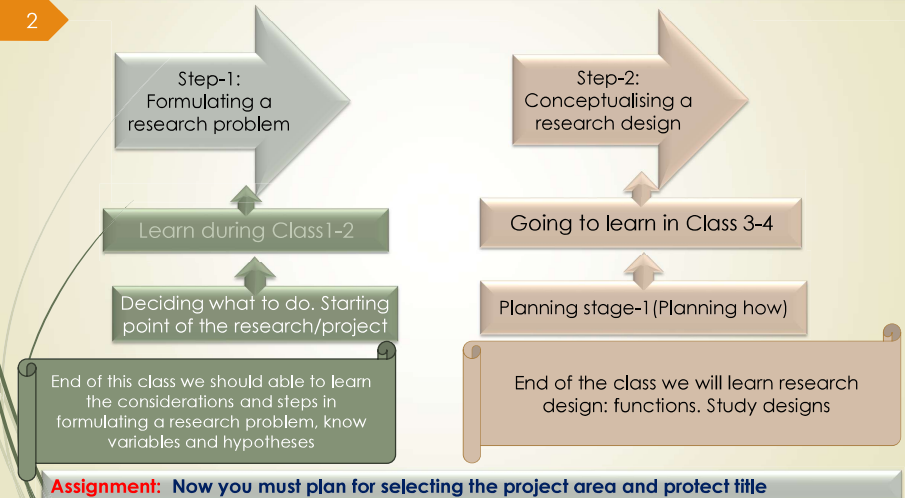
Research Methodology: B.Tech Class:3-4



Class-3: Research Designs: Exploratory and Descriptive
Class-4: Experimental Research Design

Recap from Class 1-2 and connection with class 3-4

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Class-3: Research Designs: Exploratory and Descriptive

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Learning Objective: Class-3

After the end of the class the students should be able to answer the following questions.

- What is research design and its function?
- What is the difference between research design and research methodology?
- What are the primary characteristics of a good research design?
- What is exploratory research design?
- What is descriptive research design?

The nature of research Designs

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- The very first step of research process is formulating the research problem and we learn that during class 1-2. Other way we can say it as, “**What of the study, that is the research problem**”.
- The next step of the research is , “**how of the study, e.g., the method of achieving the stated research objectives in the best possible manner**”. This we will learn in this session.

Definition of Research Design

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- A research design is the **overall plan or framework** that **guides a research project, outlining the methods and procedures used to collect, analyse, and interpret data to answer research questions or test hypotheses**.
- The above definition mentioned five important function or objective or purpose of the research design as:
 - It is an overall plan or framework
 - Guides a research project
 - Outlines the methods and procedures
 - Used to collect, analyse, and interpret data
 - Answer research questions or test hypotheses

Difference between Research Design and Research Methodology

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- **Research Design** is the framework that outlines how a study will be conducted.
- **Research Methodology** is the toolset used to implement the research design effectively.
- **The design determines what to do, while the methodology explains how to do it.**

Difference between Research Design and Research Methodology

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Research Design

- **Focus:** The overall **plan or framework** for conducting a study, including the type of research (e.g., descriptive, experimental, correlational), and the variables to be studied.
- **Purpose:** To provide a **structured approach** that ensures the data collected addresses the research problem clearly, accurately, and without bias.
- **Examples:** Descriptive research, experimental research, correlational research, diagnostic research, and explanatory research.
- **Key Considerations:** Defining the research problem, identifying variables, choosing the research approach (qualitative or quantitative), and determining the sample size and sampling techniques.

Research Methodology

- **Focus:** The specific **procedures and techniques** used to collect, analyse, and interpret data, including data collection methods, data analysis techniques, and the instruments used.
- **Purpose:** To explain the reasoning behind the **chosen approach to the research**, including the collection methods, methods of analysis, and other key points of the work.
- **Examples:** Surveys, interviews, experiments, longitudinal studies, and participant observation.
- **Key Considerations:** Data collection methods (e.g., questionnaires, interviews, observations), data analysis techniques (e.g., statistical analysis, thematic analysis), and the instruments used (e.g., questionnaires, scales).

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Formulation of research Design: Process

- To formulate the research design/process the following aspects or characteristics of research design need to be followed.
- There are five major characteristics of research design as:

- **Clarity,**
- **Validity,**
- **Reliability,**
- **Neutrality,**
- **Generalizability**



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Characteristics of a Good Research Design

Clarity:

- **Well-defined research objectives:** A good research design starts with clearly stated research questions or objectives, which guide the entire research process.
- **Precise scope:** The scope of the research should be clearly defined, specifying what the study will investigate and what it will not.
- **Concise and unambiguous language:** The research design should be expressed in a clear and understandable manner, avoiding jargon or ambiguity.

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Characteristics of a Good Research Design

Validity:

- **Accurate data collection:** The research design should employ appropriate methods for collecting data that are relevant to the research objectives.
- **Appropriate measurement tools:** The instruments used to measure variables should be valid, meaning they accurately measure what they are intended to measure.
- **Sound methodology:** The research design should be based on a sound methodology that is appropriate for the research question and type of study.

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Characteristics of a Good Research Design

Reliability:

- **Consistent results:** The research design should be able to produce consistent results if the study is repeated under similar conditions.
- **Robust data analysis:** The data analysis should be rigorous and reliable, ensuring that the evidence supports the conclusions drawn.
- **Transparent communication of results:** The research design should facilitate transparent communication of the results, allowing others to assess the validity and reliability of the findings.

Characteristics of a Good Research Design

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Neutrality:

- **Objective data analysis:** The research design should ensure that data is analyzed objectively, without bias or preconceived notions.
- **Unbiased data collection:** The data collection methods should be designed to minimize bias and ensure that all participants are treated fairly.
- **Ethical considerations:** The research design should address ethical considerations, such as informed consent, confidentiality, and the potential for harm to participants.

Characteristics of a Good Research Design

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Generalizability:

- **Applicable findings:** The research design should be designed in a way that allows the findings to be generalized to a wider population or context.
- **Representative sample:** If the study involves a sample, the sample should be representative of the population of interest.
- **Robust methodology:** The research design should be robust, ensuring that the findings are not likely to be due to chance or other confounding factors.

Classifications of the Research Designs

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- Research designs can broadly be classified into **exploratory and descriptive designs** based on the purpose of the research.
- This above classification may be too restrictive and broad way of classifying.
- However, depending on the basis of classification, other types such as experimental/casual, correlational, qualitative, quantitative, and mixed-methods designs are also identified.
- In this session we will discuss on **Exploratory and Descriptive Research Design.**

Exploratory Research Design: Definition

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- Exploratory research design is a research approach **used when the researcher has limited or insufficient knowledge about a research problem, phenomenon, or situation** and wants to explore it in order to gain a deeper understanding.
- It is **generally conducted when the problem is not clearly defined**, the important variables are not yet known, or existing theories and findings are insufficient to explain the situation.

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Exploratory Research Design: Definition



- A simple definition is: **Exploratory research is research conducted to explore an unfamiliar or insufficiently understood problem, generate insights, identify important variables, and develop questions or hypotheses for further research.**
- The definition indicates four important function/objective of the exploratory research design as:
 - **To explore an unfamiliar or insufficiently understood problem,**
 - **Generate insights,**
 - **Identify important variables,**
 - **Develop questions or hypotheses for further research.**

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Purpose of Exploratory Research Design: 1

- The main purpose is to **develop an initial understanding of a problem.**

Major purposes include:

- **1: To gain familiarity with a problem**

Exploratory research helps the researcher understand a problem about which little is known.

■ Example:

A company notices declining sales of a particular product but does not know the reason. Exploratory research can help identify possible causes.

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Purpose of Exploratory Research Design: 2

2: To identify important variables

- At the beginning of research, the researcher may not know which factors are important.

Exploratory research helps identify potential variables that can later be measured.

■ Example:

Suppose students are dissatisfied with online classes. Possible variables discovered through interviews might include:

- Internet connectivity
- Teaching quality
- Student engagement
- Assessment methods
- Lack of interaction
- Technical difficulties

These variables can subsequently be investigated through descriptive or quantitative research.

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Purpose of Exploratory Research Design: 3

3: To generate research questions

- Exploratory research can help convert a broad problem into specific research questions.

■ For example: Broad problem:

Why are customers dissatisfied with a particular banking service?

Exploration may lead to questions such as:

- **Is waiting time a problem?**
- **Are customers dissatisfied with digital services?**
- **Is staff behaviour affecting satisfaction?**
- **Are service charges considered too high?**

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Purpose of Exploratory Research Design: 4

4: To generate hypotheses

- Exploratory research can help researchers develop **tentative hypotheses** that can later be tested using quantitative or experimental research.
- For example, interviews may reveal that students who spend more time on social media have difficulty concentrating during online classes.

The researcher may subsequently formulate:

H₁: Excessive social media use is negatively associated with students' academic concentration.

The exploratory stage does not necessarily prove this hypothesis; it helps **develop a testable proposition**.

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Purpose of Exploratory Research Design: 5

5: To understand people's experiences, perceptions, and motivations

Exploratory research is particularly useful when the researcher wants to understand **why people behave in a particular way**.

- For example:

Why do some consumers prefer local brands even when international brands are available?

A survey may tell us *how many* consumers prefer local brands, but interviews may help explain **why**.

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Purpose of Exploratory Research Design: 6

6: To develop a framework or conceptual model

Exploratory research can help researchers identify relationships among different concepts and develop a preliminary conceptual framework.

- For example:

Service quality → Customer satisfaction → Customer loyalty

The researcher may initially discover these relationships through interviews and observations and later test them quantitatively.

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When Should We Use Exploratory Research Design?

Exploratory design is appropriate when:

- The research problem is **new or unfamiliar**.
- There is **limited existing literature**.
- Previous research does not adequately explain the phenomenon.
- The researcher does not yet know the important variables.
- The researcher wants to understand **why or how** something occurs.
- The researcher wants to develop research questions.
- The researcher wants to formulate hypotheses.
- A new phenomenon has recently emerged.
- Existing theories may not adequately explain the situation.

"Do I already know enough about the problem to specify exactly what I want to measure?"

If **no**, exploratory research may be appropriate.

If **yes**, you may move toward descriptive, correlational, or experimental research.

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Common Research Methods Used in Exploratory Research

- ▶ Literature Review
- ▶ Expert Interviews
- ▶ Focus Group Discussions
- ▶ Observation
- ▶ Case Study

We will discuss all about various research methods in forthcoming classes

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Case-study-3.1: "Why Does the Machine Keep Tripping?"

- ▶ Imagine a manufacturing company has a large production machine.

For several weeks, something unusual happens:

Whenever the machine operates for some time, the electrical circuit trips and the machine suddenly stops.

The maintenance team resets the circuit breaker, and the machine starts working again.

But after some time:

Trip → Reset → Machine works → Trip again

The company wants to know:

"Why is this happening?"

Problem formulation

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The problem is known, but the cause is unknown

- ▶ The company knows: **There is an electrical failure.**
- ▶ But it doesn't know the exact cause.

Several possibilities exist: Overloading, Loose electrical connection, Faulty circuit breaker, Motor overheating, Insulation problem, Voltage fluctuation, Damaged cable, Short circuit, Poor maintenance, Excessive machine load

- ▶ At this stage, nobody knows which explanation is correct.

This is where exploratory research is appropriate.

The researcher/engineer does **not immediately conclude**:

"The circuit breaker is faulty."

- ▶ That would be an assumption.

Instead:

"Let's investigate the problem and discover what might be causing it."

Justifying to why to go for Exploratory research design?

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The exploratory research question

- ▶ The researcher starts with a broad question:

"What factors may be responsible for the repeated electrical tripping of the production machine?"

- ▶ Notice that the question does not assume a particular cause.

The purpose is to **explore possible causes.**

A broad question is formulated and this will guide to proceed with proper planning towards solution in a systematic way.

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Step 1: Collect background information

- ▶ The researcher first examines existing information:
 - ▶ Machine specifications
 - ▶ Electrical diagrams
 - ▶ Maintenance records
 - ▶ Previous breakdown reports
 - ▶ Operating instructions
 - ▶ Previous repair history
 - ▶ Load requirements
 - ▶ The researcher discovers that the machine normally operates at a particular electrical load.
 - ▶ But this information alone doesn't explain the problem.
- So the investigation continues.

Information/Data
collection

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Step-2: Talk to the people involved

The researcher interviews:

▶ Machine operators

They say:

"The machine usually trips after running for about 30–40 minutes."

Another operator says:

"It happens more frequently when we operate several machines simultaneously."

Interesting.

▶ Maintenance technicians

One technician says:

"We replaced the circuit breaker last month, but the problem continued."

Another says:

"The motor becomes quite hot before the trip occurs."

Now the researcher has new clues.

Research
methods are
adopted for
information/data

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Step 3: Observe the situation

- ▶ The researcher observes the machine during operation.
- After 30 minutes:
Motor temperature begins to increase.
- After 40 minutes:
The electrical current rises.
- A few minutes later:
- ▶ **Circuit breaker trips.**
- The researcher records these observations.
- Now a possible relationship is emerging:
- Longer operation → Motor temperature increases → Current increases → Circuit trips**
- But this is still only an initial explanation.**
- The researcher has not yet established the exact cause.**

Another research
method that is
observation is adopted
for information/data

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Step 4: Explore different possible explanations

- ▶ The researcher develops several possibilities.
- Possibility 1:** The machine is drawing excessive current.
- Possibility 2:** The motor is overheating.
- Possibility 3:** There is a problem with the electrical supply.
- Possibility 4:** The circuit breaker is incorrectly rated.
- Possibility 5:** There is a mechanical problem causing the motor to work harder.
- ▶ Notice something important:
- Exploratory research does not start by choosing one explanation.**
- It keeps multiple possibilities open.

Analysis of
information/data
has been done
with
hypothetical/pos-
sible solutions

Step 5: Examine the evidence

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The researcher collects information about:

- Current levels
- Voltage levels
- Motor temperature
- Machine operating time
- Machine load
- Frequency of tripping
- Maintenance history
- Environmental conditions

Suppose the researcher notices:

Condition	Observation
Machine starts	Normal
After 10 minutes	Normal
After 20 minutes	Current begins increasing
After 30 minutes	Motor temperature rises
After 40 minutes	High current
Shortly afterward	Circuit trips

Data analysis has been done here

This pattern provides an important clue.

Step 6: Talk to an electrical engineer

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The researcher consults an electrical engineer.

The engineer suggests:

"The increasing current and temperature may indicate that the motor is experiencing excessive load or another fault. We need further diagnostic testing to determine the actual cause."

This is an important distinction.

The exploratory stage has **identified possible explanations**, but it has not necessarily proven the cause.

Information/data are analysed in the previous step and now we try to interpret from the analysis for a solution to the problem

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Step 7: The cause of the problem becomes clearer

Initially:

"The machine keeps tripping. We don't know why."

After exploration:

"The tripping appears to be associated with increased motor temperature and current after prolonged operation. Possible contributing factors include excessive mechanical load, motor condition, or electrical supply issues."

Now the researcher finds a solution to the problem.

Finalize the solution with confirmation

What next?

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Generally one research design may not completely leads too final result or complete the research process.

For this problem here exploratory research ends. However, there is a need to follow other research designs for further research. The reason to follow various other research designs are as follows:

Stage 1: Exploratory

"What could be causing the problem?"

The researcher investigates broadly.

Stage 2: Descriptive/Diagnostic

"What exactly is happening, and how frequently?"

The researcher measures: current, voltage, temperature, load, frequency of trips, etc.

Stage 3: Analytical/Correlational

"Are the variables related?"

For example: Is increased machine load associated with increased current?

Stage 4: Experimental/Testing

"If we change the suspected factor, does the problem change?"

For example, under appropriate engineering safety procedures, a controlled test might examine whether reducing the machine's operating load changes the frequency of trips.

Complex problems very often need multiple research designs to follow

Descriptive Research Design: Definition

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- Descriptive research design is a **systematic, non-experimental methodology** used to accurately and systematically describe a population, situation, or phenomenon.
- It answers the **"what," "who," "where," and "when"** questions of a research problem, but **not the "why" or "how."**
- In this design, the researcher observes and measures variables as they naturally occur without manipulating or controlling any of them.
- It provides a "snapshot" of a specific situation at a given point in time (cross-sectional) or tracks changes over a period (longitudinal).

Purpose of Descriptive Research Design

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The main purpose of descriptive research design is to provide an accurate and detailed description of a population, event, or phenomenon. Its important purposes include:

1) To Describe Existing Conditions

Descriptive research helps researchers understand the current status of a situation.

Example: A researcher may study the level of digital literacy among rural students to understand their present knowledge and skills.

2) To Identify Characteristics of a Population

It helps in understanding demographic, social, economic, or behavioral characteristics.

Example: A survey may identify the age, income, education level, and purchasing habits of customers.

Continue...

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3) To Measure Frequency and Trends

Descriptive research identifies how often something occurs and detects patterns over time.

Example: A study may measure the prevalence of online learning among university students.

4) To Provide Baseline Information

The findings of descriptive studies can serve as a foundation for further research.

Example: A health survey may provide baseline data before introducing a public health program.

5) To Support Decision-Making and Planning

Organizations and policymakers use descriptive research findings for planning and policy development.

Example: Government agencies may conduct population surveys to plan healthcare or education programs.

6) To Compare Groups or Conditions

Descriptive research can compare differences between groups.

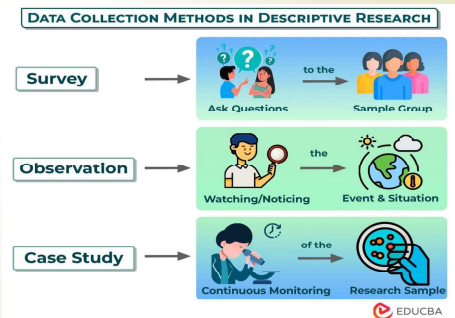
Example: Comparing job satisfaction levels between public and private sector employees.

Common Research Methods Used

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Descriptive research relies on several data collection methods, which can be quantitative, qualitative, or mixed. The most common include:

- Surveys and Questionnaires**
- Observational Studies**
- Case Studies**
- Focus Groups**
- Demographic/Census Analysis**



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When Should We Use Descriptive Research Design?

- **When the Research Problem is Clearly Defined**
- **When You Need to Answer "What," "Who," "Where," or "When" Questions**
 - When You Need to Establish a Baseline or Benchmark
 - When You Want to Identify Patterns, Trends, or Frequencies
 - When You Need to Profile a Population or Segment
 - When Variables Cannot Be Manipulated
- **When You Need Data for Decision-Making**
 - When You Want to Compare Groups or Categories
- **When Resources Are Limited for Experimental Research**
- **When You're Conducting Preliminary Research for Future Studies**

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Difference between the exploratory and descriptive research design

1. Purpose and Outcome

- **Exploratory:** The goal is to *explore*. It is used when the researcher is navigating uncharted territory. The outcome is not a final answer, but rather a set of hypotheses, refined research questions, or a clearer definition of the problem.
- **Descriptive:** The goal is to *describe*. It is used when the researcher knows exactly what they are looking for. The outcome is a factual, statistical, or detailed account of the current state of affairs.

2. Level of Structure

- **Exploratory:** *Unstructured*. An interviewer might change their questions mid-conversation based on the participant's responses. The research design evolves as data is collected.
- **Descriptive:** *Highly structured*. A survey questionnaire is finalized before data collection begins. Every participant is asked the exact same questions in the same order to ensure consistency and reliability.

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Difference between the exploratory and descriptive research design

3. Sampling Strategy

- **Exploratory:** Uses *non-probability sampling* (e.g., convenience sampling, snowball sampling, purposive sampling). The goal is to find people who have rich information, not to represent the whole population. (e.g., interviewing 5 industry experts).
- **Descriptive:** Uses *probability sampling* (e.g., simple random sampling, stratified sampling). The goal is to ensure the sample accurately reflects the broader population so that findings can be generalized. (e.g., surveying 1,000 randomly selected customers).

4. Relationship to Hypotheses

- **Exploratory:** *Generates hypotheses*. (e.g., "Based on interviews, we hypothesize that poor customer service is causing the drop in sales.")
- **Descriptive:** *Describes or tests the prevalence of those hypotheses*. (e.g., "A survey reveals that 65% of customers cite poor customer service as their primary reason for leaving.")

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Case-study- 3.2: Mobile banking app

- **The Problem:** A popular mobile banking app is experiencing a high rate of unexpected crashes, but the development team doesn't know where to start debugging.
- **The Descriptive Research Goal:** To accurately describe *what* is crashing, *where* (which devices/OS), and *when* it happens, without trying to prove *why* it happens.

Step-by-Step Approach:

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- **Step-1: Data Collection:** Extract 30 days of automated crash logs from the app's monitoring tool (e.g., Firebase Crashlytics). No code is changed or tested.
- **Step-2: Variables Measured:** Device model, Operating System (OS) version, app feature being used at the time of crash, and time of day.
- **Step-3: Descriptive Analysis:** Calculate frequencies and percentages of the logged data.
- **Step-4: The Findings:** The data reveals that 72% of all crashes occur exclusively on Android OS version 13, specifically when users attempt to use the "Scan Check" feature, mostly between 5:00 PM and 8:00 PM. iOS users and other features show a near-zero crash rate.

How it Solves the Problem

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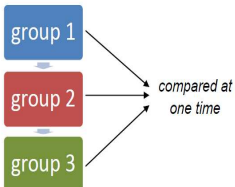
- Instead of wasting weeks blindly reviewing the entire app's codebase or running expensive A/B tests, the engineering team now has a precise, factual target. They can immediately assign their best Android developer to investigate the "Scan Check" module for OS 13 compatibility issues.
- (Note: **This is purely descriptive. It tells the team exactly what and where the problem is, providing the necessary baseline to fix it, without claiming to have experimentally proven the root cause**).

Cross-Sectional Study

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Cross-Sectional Research Study:

different groups



Definition:

A **cross-sectional study** is a type of research design in which data is collected from a population or sample at **one specific point in time**. It provides a "snapshot" of the characteristics, behaviors, attitudes, or conditions of a group at a particular moment.

In a cross-sectional study, different individuals or groups are studied simultaneously, but researchers do not follow the participants over time.

Example:

A researcher conducts a survey in 2026 to study the level of stress among university students. Data is collected once from students during that academic year.

Purpose:

- To measure the current status of a phenomenon.
- To estimate the prevalence or frequency of a condition.
- To compare differences between groups at a particular time.

Longitudinal Study

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Definition: A **longitudinal study** is a research design in which data is collected from the **same participants or population repeatedly over an extended period of time**. It examines changes, developments, or trends over time.

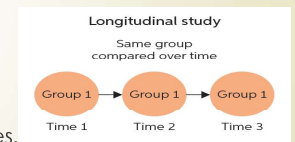
Researchers observe participants at multiple points to understand how variables change and develop.

Example:

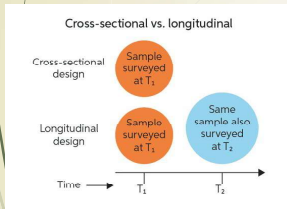
A researcher studies the career development of the same group of university graduates every year for ten years.

Purpose:

- To study patterns of change over time.
- To understand development, growth, or decline.
- To identify long-term relationships between variables.



Difference Between Cross-Sectional and Longitudinal Studies

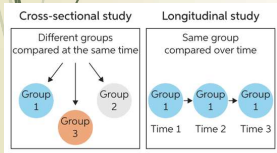


- **Cross-sectional studies** examine **different people at one point in time** and are useful for understanding the current status of a phenomenon.
- **Longitudinal studies** examine **the same people repeatedly over time** and are useful for understanding changes, development, and long-term patterns.

In simple terms:

Cross-sectional = "A snapshot of a situation at one time."

Longitudinal = "A movie showing how a situation changes over time."



Class-4: Experimental Research Design

Learning Objective: Class-4

After the end of the class the students should be able to answer the following questions.

- How to define experiment and explain the concept of causality.
- What are the necessary conditions for drawing causal inferences?
- What is internal and external validity of the experiment?
- What are the methods to control the extraneous variables?
- What are the various types of experimental design?

What is an experiment?

- Experimental research design is used to **examine cause-and-effect relationships between variables (known as causality)**.
- It involves **deliberately changing or manipulating one variable (called the independent variable) and observing its effect on another variable (called the dependent variable) under controlled conditions**.
- The main idea behind experimental research is: **"If we change one factor, does it produce a change in another factor?"**
- **For example:** A researcher wants to know whether a new teaching method improves students' academic performance. The researcher introduces the new teaching method to one group of students and compares their performance with another group taught using the traditional method.

Definition of experimental design

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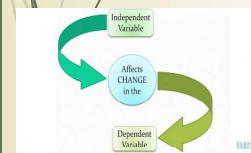
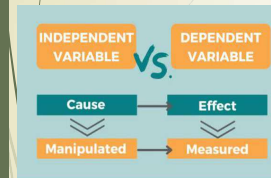
Experimental research design is a systematic research approach in which the researcher manipulates one or more independent variables, controls other factors, and observes the impact on dependent variables to determine causal relationships.

- In this definition three important factor an experiment involves are emphasized:
 - **Manipulation:** changing or introducing an independent variable.
 - **Control:** maintaining control over other influencing factors.
 - **Observation/Measurement:** measuring the effect on the dependent variable.

Necessary conditions for making causal inferences

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The following three are the necessary conditions for making causal inferences:



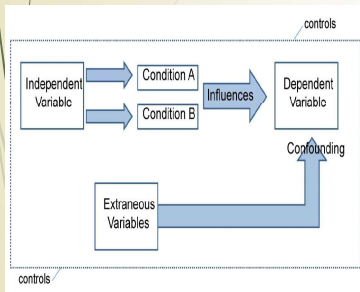
- **1: Concomitant variations:** This refers to the extent to a cause (independent variable, e.g. input/control variable) and effect (on the dependent variable e.g. output) **occur together or vary together**. This relation or variation regulates the experiment objective to be proved or justified.
- **2: Time order of occurrence of variables:** This condition means that the **causal variable (that means the cause/independent variables) must occur prior to or simultaneously with the effect/dependent/output variable**. This condition must require because the purpose for which the experiment need to do will not be fruitful.

Continue...

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■ 3: Absence of other possible causal factors:

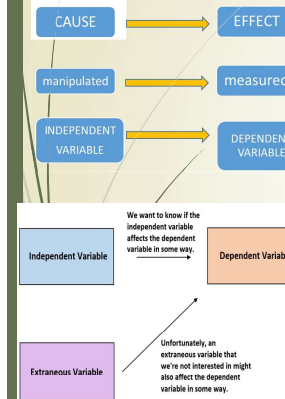
Generally, the experiments we do with an objective to investigate certain cause and effect relationship between the independent/input and dependent/output variables. However, these are controlled variables and in real-time conditions there **may be certain uncontrolled or extraneous variables may occur** and their impact cannot be ignored. With an assumption or condition of these causal factors we do experiment if not consider in our study.



Five major Concepts used in experiments

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- **Independent variables:** These variables are also known as explanatory variables or treatments or controlled input variables. Example: Type of teaching method (traditional vs. digital learning).
- **Test units:** Test units are those entities on which treatments/independent variables are applied.
- **Dependent variables:** These variables measure the effect of treatments (independent variables) on the test units. Example: Students' examination scores.
- **Experiment:** An experiment is executed when the researcher manipulates one or more independent variables and measures their effect on the dependent variables while controlling the effects of the extraneous variables.
- **Extraneous variables:** These are the variables other than the independent variables which influence the response of test units to treatments.

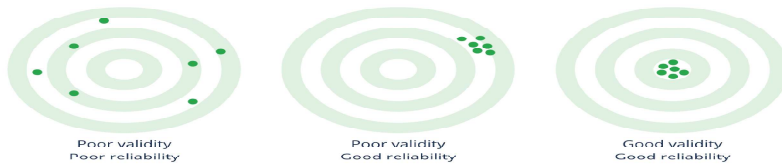


Validity in the experimentation

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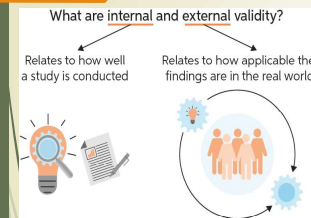
- Validity refers to the extent to which an experiment actually measures or tests what it is intended to measure or test.
- In simple terms, an experiment is valid if its results are accurate and meaningful for the research question.

Reliability vs validity



Internal and external validity

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Internal Validity:

- It shows whether the observed effect was actually caused by the independent variable rather than other factors.
- Example:** If a study tests whether a new teaching method improves exam scores, internal validity is high if factors such as students' prior knowledge, study time, and teacher differences are controlled.

External Validity:

It refers to whether the findings can be **generalized to other people, settings, or situations.**

- Example:** Results from a study conducted with 20 university students may have limited external validity if we want to apply them to all students.

Internal validity asks:
"Did the treatment really cause the effect?"

External validity asks:
"Can the results be applied beyond this particular experiment?"

Internal Validity

External Validity

Controlling extraneous variables

Controlling for environmental variables

Difficulty generalizing to the real world

Internal Validity

External Validity

Too many confounding variables

Real world conditions

Able to generalize to the real world

BALANCING INTERNAL AND EXTERNAL VALIDITY

Often, studies stricter internal validity (more criteria) leads to reduced external validity. We generally put many assumptions and neglect many factors while doing experiment and that leads to less external validity and away from reality.

Factors affecting internal validity of the experiment

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- History:** Events occurring during the experiment, apart from the treatment, may influence the results.
Example: A major event happens during a study on students' stress levels.
- Maturation:** Participants naturally change over time due to age, fatigue, learning, development, or experience.
Example: Children improve their skills simply because they grow older during a long study.
- Testing Effect:** Taking a test once can influence performance on a later test.
Example: Participants score higher on a post-test because they remember questions from the pre-test.

Continue...

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- **Instrumentation:** Changes in measurement tools, procedures, or observers can affect the results.
Example: Different observers use different standards to rate participants.
- **Selection Bias:** Participants in different groups may differ systematically before the experiment begins.
Example: The treatment group contains more highly motivated students than the control group.
- **Regression to the Mean:** Extremely high or low scores tend to move closer to the average when measured again.
- **Experimental Mortality (Attrition):** Participants dropping out of the study can change the composition of the sample and affect results.

Factors affecting the external validity

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- 1: Sampling Bias:** If the sample does not represent the target population, the results may not be generalizable.
Example: Studying only urban students and applying the findings to all students.
- 2: Sample Characteristics:** Age, gender, education, socioeconomic background, culture, and other characteristics of participants can affect generalizability.
Example: Results from adults may not apply to children.
- 3: Interaction of Selection and Treatment:** The treatment may work differently for different types of participants.
Example: A teaching method may work well for high-achieving students but not for students with learning difficulties.
- 4: Interaction of Setting and Treatment:** Results obtained in one environment may not apply to other environments.
Example: A teaching technique successful in a well-equipped private school may not produce the same results in a rural school with limited resources.

Continue....

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- 5: Interaction of History and Treatment:** Results may depend on the particular time or circumstances in which the experiment was conducted.
Example: A stress-management program tested during an examination period may produce different results at another time.
- 6: Artificial Experimental Conditions:** Highly controlled laboratory conditions may differ from real-life situations.
Participants may behave differently when they know they are being observed.
- 7: Hawthorne Effect:** Participants may change their behaviour simply because they know they are part of a study.
This can make it difficult to generalize the findings to normal situations.
- 8: Pre-test Sensitization:** Taking a pre-test may make participants more aware of the treatment or research topic, causing results that may not occur in people who did not take the pre-test.

Methods to control extraneous variables: 1- Randomization

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1. Randomization

- **Definition:** Randomization involves assigning participants to experimental and control groups by chance.
- **How it controls extraneous variables:** Random assignment ensures that individual differences (such as intelligence, motivation, age, or ability) are distributed equally among groups.
- **Example:** In a drug experiment, participants are randomly assigned to:
 - Treatment group
 - Control group

This reduces the possibility that one group is naturally healthier than the other.

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Methods to control extraneous variables: 2- Matching

2: Matching

- **Definition:** Matching involves selecting participants with similar characteristics and assigning them to different groups.
- **How it controls extraneous variables:** Participants are matched based on variables that may influence the outcome.
- **Example:** In an education study, students may be matched based on:
 - Age
 - Previous grades
 - Intelligence level

Then matched participants are divided into experimental and control groups.

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Methods to control extraneous variables: 3- Holding Variables Constant (Constancy of Conditions)

3: Holding Variables Constant (Constancy of Conditions)

- **Definition:** The researcher keeps certain conditions the same for all participants.
- **How it controls extraneous variables:** It prevents environmental differences from affecting the results.
- **Example:** In a learning experiment:
 - Same classroom
 - Same teacher
 - Same duration
 - Same instructions

are provided to all participants.

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Methods to control extraneous variables: 4- Statistical Control

4: Statistical Control

- **Definition:** Statistical control involves using statistical techniques to remove the influence of extraneous variables during data analysis.
- **Common techniques:**
 - Analysis of covariance (ANCOVA)
 - Regression analysis
 - Partial correlation
- **Example:** A researcher studying the effect of training on employee performance controls for previous work experience statistically.

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Environments of conducting experiment

- The environment in which an experiment is conducted plays a massive role in shaping its validity.
- Researchers typically choose between two primary settings: **laboratory experiments** and **field experiments**.
- Each environment creates a different balance between control and real-world applicability.

1. Laboratory Experiments

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A laboratory experiment is conducted in a highly controlled, artificial environment created specifically for research (such as a university lab, a soundproof room, or a computer testing station).

- **How it works:** The researcher manipulates the independent variable while actively stripping away, isolating, or holding constant as many extraneous variables as possible.

Strengths:

- **High Internal Validity:** Because the environment is strictly controlled, researchers can confidently rule out alternative explanations and establish clear cause-and-effect relationships.
- **Replicability:** Standardized procedures make it easy for other researchers to repeat the study exactly.

Weaknesses:

- **Low External Validity:** The artificial setting can make participants behave differently than they would in the real world (sometimes called the Hawthorne effect or demand characteristics).



2. Field Experiments

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A field experiment is conducted in a natural, real-world setting where participants go about their everyday lives (such as a grocery store, a workplace, a classroom, or a busy street).

- **How it works:** The researcher still manipulates the independent variable, but they do so out in the wild, without altering the natural environment.

Strengths:

- **High External Validity (Ecological Validity):** Because the setting is real, the results reflect genuine human behaviour and can easily be generalized to everyday life.
- **Less Awareness:** Participants often do not realize they are part of an experiment, reducing evaluation apprehension and artificial behavior.

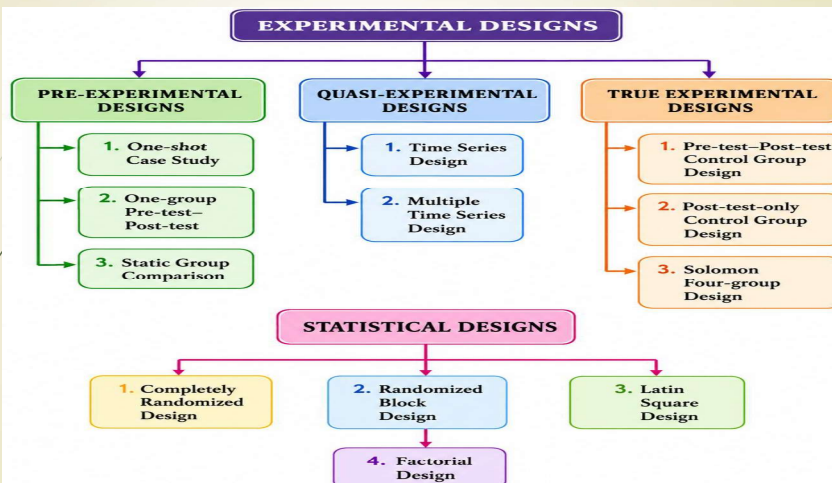
Weaknesses:

- **Low Internal Validity:** It is much harder to control for confounding variables, history, and extraneous events in a chaotic, real-world environment.
- **Ethical and Practical Challenges:** Gaining consent can be difficult, and researchers have less control over disruptions or participant dropouts.



A classification of experimental designs

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1: Pre-Experimental Designs

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Pre-experimental designs are basic experimental designs in which the researcher has **little or no control over extraneous variables** and generally does not use random assignment. They are useful for preliminary or exploratory studies.

Types & Examples:

1. One-shot Case Study

Example: A teacher introduces a new teaching method to one class and tests the students only at the end.

2. One-group Pre-test-Post-test

Example: Students take a test before and after a new teaching method to see whether their scores improve.

3. Static Group Comparison

Example: One class receives a new teaching method while another class does not, and their final test scores are compared.

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2: Quasi-Experimental Designs

Quasi-experimental designs are used when the researcher can control the **timing of measurements and treatment**, but cannot completely randomly assign participants to groups.

Types & Examples:

1. Time Series Design

Example: A company records employee productivity for several months, introduces a training programme, and continues recording productivity afterward.

2. Multiple Time Series Design

Example: Productivity is measured repeatedly before and after training for both a trained group and an untrained comparison group.

Key point: Useful when random assignment is difficult or impossible.

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3: True Experimental Designs

True experimental designs involve **random assignment of subjects to experimental and control groups**, allowing the researcher to establish stronger cause-and-effect relationships.

Types & Examples:

1. Pre-test-Post-test Control Group Design

Example: Students are randomly divided into two groups. Both take a pre-test; one receives a new teaching method; both take a post-test.

2. Post-test-only Control Group Design

Example: Employees are randomly divided into training and control groups. After training, their performance is compared without giving a pre-test.

3. Solomon Four-Group Design

Example: Participants are divided into four groups to determine both the effect of a treatment and whether taking a pre-test itself affects the results.

Key point: Randomization provides stronger control and improves internal validity.

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4: Statistical Designs

Statistical designs use **statistical principles to control variation and study the effects of one or more independent variables** efficiently.

Types & Examples:

1. Completely Randomized Design

Example: 30 students are randomly assigned to three different teaching methods.

2. Randomized Block Design

Example: Students are first grouped according to ability level and then randomly assigned to different teaching methods within each level.

3. Latin Square Design

Example: Three advertisements are tested across three different cities and three different time periods, while controlling for both city and time effects.

4. Factorial Design

Example: A researcher studies the effects of **teaching method and study time** simultaneously on students' performance.

Key point: Useful when researchers need to control variation or study multiple factors simultaneously.

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Short Story: The Mystery of the Vanishing Plants – A Research Design Exploration

- Lena, a young environmental scientist, loved exploring nature.
- One day, while hiking near her town, she noticed something strange: several rare plants that used to thrive in the area had started disappearing.
- Determined to solve this mystery, she decided to conduct research.

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Step 1: Choosing the Research Design

Lena knew she had to carefully plan her study, so she considered different research designs:

- **Exploratory Research:** If she knew nothing about why the plants were disappearing, she could start by gathering initial information from books, scientists, and local farmers.
- **Descriptive Research:** If she wanted to document how many plants had vanished and where, she could conduct a survey and create a map.
- **Experimental Research:** If she suspected that pollution from a nearby factory was causing the disappearance, she could design an experiment to test how different levels of pollution affect plant growth.
- **Correlational Research:** If she wanted to check whether the plant decline was linked to temperature changes, she could analyze historical temperature records and compare them to plant disappearance patterns.

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Step 2: Conducting the Research

- Lena decided to use a **descriptive research design** first to document the affected areas. She visited different locations, counted the remaining plants, and recorded environmental conditions like soil quality, water availability, and air pollution levels.
- After collecting enough data, she noticed a pattern, plants near the factory were disappearing faster than others. To confirm this, she moved to **experimental research** by growing the same plant species in two environments: one with clean soil and another with soil from the polluted area.

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Step 3: Analysing and Concluding

- After weeks of observation, Lena saw that plants in polluted soil grew weak and died faster.
- Her findings confirmed that pollution was a major factor.
- She presented her research to local authorities, and thanks to her structured research design, they took action to reduce pollution.
- Lena's mystery was solved, and her research design had helped uncover the truth!
- **This story illustrates how different research designs help in different situations.**

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Thanks